ChE 210 Homework Set #4 Due on Tuesday, 11 February 2020

(1) Seawater containing 3.5 wt% salt passes through a series of 10 evaporators. Roughly equal quantities of water are vaporized in each of the 10 units and then condensed and combined to obtain a product stream of fresh water. The brine leaving each evaporator (except for the tenth and final unit) is fed into the next evaporator. The brine leaving the tenth evaporator contains 5 wt% salt.

(a) Draw a flowchart of the process. Label all the necessary streams. Can you draw your flow diagram in such a way as not to need to show all ten units explicitly?

(b) Write in order the set of equations you would solve to determine the fractional yield of fresh water from the process (kg H2O recovered/kg H2O in feed) and the weight percent of salt in the solution leaving the fourth evaporator. Each equation you write should contain no more than one previously undetermined variable. In each equation, circle the variable for which you would solve.

(c) Solve the equations derived in part (b) for the two specified quantities.

(2) Biological organisms need and environment of humid air enriched in oxygen in order to grow. To provide a continuous steam of appropriate humid air, the following three input streams are fed into an evaporator unit.

(1) Air (21 mol% O_2, the balance N_2)
(2) Liquid water, fed at a rate of 20.0 cm^3/min
(3) Pure oxygen, with a molar flow rate one fifth of the molar flow rate of the air input.

The single, gas output stream contains 1.5 mole % water.

(a) Draw and label a flow diagram of the process. Label the total molar flow rates of the air, water, and pure oxygen input streams as $\dot{n}_1$, $\dot{n}_2$, and $\dot{n}_3$, respectively. For the exit stream label the total molar flow rate as $\dot{n}_4$ and the mole fractions of oxygen, nitrogen, and water as $x_{4,O}$, $x_{4,N}$, and $x_{4,W}$, respectively.
(b) Do a degree of freedom analysis.
(c) Calculate all unknown stream variables.

Data for water: $M_{H_2O} = 18.015$ g/mol and density 1.00 g/cm^3.

(3) Murphy 2.66

(4) Murphy 2.69 Although the problem statement says you do not have to solve the equations, please solve them anyway. You are free to use a numerical method or software package.